FORCES AND MOTION

GUIDED NOTES · PHASE 02 · 6 SECTIONS · TEACHER KEY

NAME _____ DATE _____ PERIOD _____

UNIT LEARNING TARGETS

- I can state Newton's three laws and use Fg = mg to find an object's weight.
- I can draw a free-body diagram and find the net force on an object, including forces at an angle.
- I can split an object's weight into parts along and into a ramp and find the normal force.
- I can use ∑F = ma to find acceleration, including on a ramp and in an elevator.
- I can tell static from kinetic friction and find friction, net force and acceleration on level ground and on a ramp.
- I can find tension and acceleration for objects joined by cables, ropes and pulleys.

KEY TERMS

- **S03.1** Force, newton (N), mass, weight, equilibrium, net force
- **S03.2** Free-body diagram (FBD), normal force, tension, resultant, components
- **S03.3** Incline, Fg,parallel, Fg,perp
- **S03.4** Acceleration, apparent weight
- **S03.5** Static friction, kinetic friction, coefficient of friction (μ)
- **S03.6** Ideal string, pulley, Atwood machine

HOW THESE NOTES WORK

- The left column names the slide each block comes from.
- While that slide is up, write each idea in your own words on the lines under its prompt.
- In each work box, use the positive direction given, or choose and state one, and box your final answer with units and direction.

SECTIONS IN THIS UNIT

- ☐ S03.1 Newton's Laws
- ☐ S03.2 Free-Body Diagrams
- ☐ S03.3 Forces on an Incline
- ☐ S03.4 Net Force and Acceleration
- ☐ S03.5 Friction
- ☐ S03.6 Connected Objects

EQUATION TOOLBOX

Newton's laws

$Fg = mg$

$\sum F = ma$

$a = \frac{\sum F}{m}$

$FN - mg = ma$ (up is positive)

Components

$Fx = F\cos\theta$

$Fy = F\sin\theta$

$\sum F = \sqrt{((\sum Fx)^2 + (\sum Fy)^2)}$

$\theta = \tan^{-1}(\sum Fy / \sum Fx)$

Ramps and friction

$Fg,parallel = mg\sin\theta$

$Fg,perp = mg\cos\theta$

$FN = mg\cos\theta$ (ramp, no other perpendicular force)

$fs,max = \mu_s FN$

$fk \approx \mu_k FN$

∑F = net force (sum of all forces).
g = 9.8 m/s² on Earth. 1 N = 1 kg·m/s². μ has no units. Frictionless ramp: a = g sin θ.

Newton’s Laws

LEARNING TARGET I can state Newton’s three laws and use Fg = mg to find an object’s weight.

FORCE · MASS AND WEIGHT

Slides 3–4

Common mix-up: using mass and weight as if they were the same thing.

Force: definition and unit

A force is a push or a pull. It is a vector, with a size and a direction, measured in newtons (N).

Mass vs. weight: meaning and units

Mass: how much matter, in kg; stays the same when you travel. Weight: the force of gravity, in N; changes when gravity changes.

Weight: equation, g on Earth, and the 10 kg example

Fg = mg, with g = 9.8 m/s² on Earth. 10 kg: 98 N down on Earth, 16 N down on the Moon.

WEIGHT ON EARTH AND THE MOON

Slide 5

Same mass, different gravity

Given: m = 60 kg; Earth: g = 9.8 m/s²; Moon: g = 1.6 m/s². **Find:** weight in each location.

WORK / show your reasoning and units

Fg = mg

Earth: Fg = 60(9.8) = 588 N down

Moon: Fg = 60(1.6) = 96 N down

Same mass. Different gravity. Different weight.

Answer: Earth: Fg = 588 N down Moon: Fg = 96 N down

NEWTON'S THREE LAWS

Slides 6–8

Common mix-up: thinking motion needs a net force. Watch the puck on slide 6.

Newton's 1st law

An object at rest stays at rest, and an object in motion keeps a constant velocity, unless a net external force acts on it.

Equilibrium: condition and the two possibilities

Forces balance, ∑F = 0. The object stays at rest, or moves at constant speed in a straight line.

Newton's 2nd law: equation and meaning

∑F = ma, so a = ∑F / m. Net force, ∑F, is the sum of all forces acting on the object. Same mass: more net force, more acceleration. Same force, twice the mass: half the acceleration.

Newton's 3rd law, and why the pair does not cancel

For every action, there is an equal and opposite reaction. The pair acts on two objects, so it does not cancel on one object's FBD.

RECALL

SECTION SUMMARY

Close your notes before you write.

Use the checklist to find what still needs practice.

State Newton's three laws in your own words, the condition for equilibrium, and the equation for weight.

1st: ∑F = 0, no change in velocity. 2nd: ∑F = ma. 3rd: equal, opposite pairs. Equilibrium: ∑F = 0. Fg = mg.

- [ ] I can tell mass (kg) from weight (N) and find weight with Fg = mg.
- [ ] I can state all three laws and explain equilibrium.

Free-Body Diagrams

LEARNING TARGET I can draw a free-body diagram and find the net force on an object, including forces at an angle.

FREE-BODY DIAGRAMS · NORMAL FORCE

Slides 10–11

Common mix-up: assuming FN always equals the weight.

Drawing an FBD: the rules

Box or dot; one labeled arrow per force; longer arrow = larger force; no motion arrows; only forces on the chosen object.

Common forces: Fg, FN, T

Fg: gravity pulls down. FN: surface pushes perpendicular. T: rope pulls along the rope.

Normal force: direction, and when FN = Fg

Perpendicular push from the surface: up on a floor, tilted on a ramp. FN = Fg only on a level surface, no other vertical forces, no vertical acceleration.

BALANCED FORCES: BOOK AND LAMP

Slide 12

Two FBDs: book and lamp

Given: 2 kg book at rest on a table; 1.5 kg lamp at rest on a rope; $g = 9.8\ m/s^2$.

Find: the free-body diagram of each, and the upward force on each.

WORK / show your reasoning and units

Book FBD: FN up, Fg down, equal arrows. Lamp FBD: T up, Fg down, equal arrows.

Book: FN = Fg = 2(9.8) = 19.6 N up

Lamp: T = Fg = 1.5(9.8) = 14.7 N up

Answer: Book: FN = 19.6 N up Lamp: T = 14.7 N up

NET FORCE IN ONE DIMENSION

Slide 13

Slide 14: worked on the board.

Given: crate; 30 N right; 10 N left. Positive: right. **Find:** ∑F and its direction; write the method first.

WORK / show your reasoning and units

Method: choose a positive direction, add forces with signs, x and y separately.

Right: +30 N Left: –10 N

∑F = 30 – 10 = 20 N

20 N to the right

Answer: ∑F = 20 N to the right

RESULTANT FORCE AND DIRECTION

Slide 15

Given: 40 N right (+x) and 30 N up, perpendicular. **Find:** the size and direction of the resultant.

WORK / show your reasoning and units

$\sum F = \sqrt{(40^2 + 30^2)} = 50$ N

$\theta = \tan^{-1}(30/40) = 36.9°$

50 N, 36.9° above +x

Answer: $\sum F$ = 50 N at θ = 36.9° above +x

BREAKING A FORCE INTO COMPONENTS

Slide 16

Given: F = 100 N pulling 30° above the horizontal. **Find:** Fx and Fy, from the component equations.

WORK / show your reasoning and units

$Fx = F \cos\theta = 100 \cos 30° = 86.6$ N horizontal, toward the pull

$Fy = F \sin\theta = 100 \sin 30° = 50.0$ N up

Check: $\sqrt{(86.6^2 + 50.0^2)} \approx 100$ N

Answer: Fx = 86.6 N horizontal, toward the pull Fy = 50.0 N up

TWO ANGLED FORCES: TUGBOATS

Slide 18

Check the quadrant before reporting an angle

Slide 17: worked on the board.

Given: top view; 500 N at 30° above +x; 400 N at 45° above –x. Positive: +x right, +y up the page.

Find: the net force on the barge and its direction.

WORK / show your reasoning and units

$\sum Fx = 500 \cos 30° - 400 \cos 45° = 150$ N

$\sum Fy = 500 \sin 30° + 400 \sin 45° = 533$ N

$\sum F = \sqrt{(150^2 + 533^2)} \approx 554$ N

$\theta = \tan^{-1}(533/150) \approx 74°$

Answer: $\sum F \approx$ 554 N at $\theta \approx$ 74° above +x

RECALL

SECTION SUMMARY

Close your notes before you write.

Use the checklist to find what still needs practice.

Summarize how to find the net force from a free-body diagram, including a force at an angle.

One arrow per force; choose a positive direction. Split angled forces with Fx = F cos θ and Fy = F sin θ, add each direction with signs, then combine with $\sqrt{((\sum Fx)^2 + (\sum Fy)^2)}$.

- [ ] I can draw an FBD with one labeled arrow per force.
- [ ] I can add forces with signs and split an angled force into components.

Forces on an Incline

LEARNING TARGET I can split an object's weight into parts along and into a ramp and find the normal force.

GRAVITY ON A RAMP · NORMAL FORCE ON AN INCLINE

Slides 20–21

Common mix-up: assuming FN equals the weight on a ramp.

Ramp axes and the weight components

One axis along the ramp, one perpendicular. Fg,parallel = mg sin θ down the slope; Fg,perp = mg cos θ into the surface.

FN on a ramp: equation, condition, steepness

FN = mg cos θ (no extra perpendicular force, no perpendicular acceleration). Steeper ramp, less support: 15°: 0.97mg; 60°: 0.50mg.

INCLINE EXAMPLE: A 5 KG BLOCK

Slide 22

Given: m = 5 kg; frictionless 30° ramp; g = 9.8 m/s². **Find:** FBD, Fg,parallel, Fg,perp, and FN.

WORK / show your reasoning and units

FBD: Fg down, FN perpendicular out of the ramp; Fg split into Fg,parallel and Fg,perp.

Fg = mg = 5(9.8) = 49 N

Fg,parallel = 49 sin 30° = 24.5 N down the ramp

Fg,perp = 49 cos 30° = 42.4 N into the ramp

FN = 42.4 N, perpendicular out of the ramp

Answer: Fg,parallel = 24.5 N down the ramp Fg,perp = 42.4 N into the ramp FN = 42.4 N out of the ramp

INCLINE EXAMPLE: A PUSH UP THE RAMP

Slide 23

Given: m = 12 kg; frictionless 25° ramp; 30 N push uphill. Positive: downhill. **Find:** Fg,parallel and ∑F.

WORK / show your reasoning and units

Fg,parallel = 12(9.8) sin 25° = 49.7 N downhill

∑F = 49.7 – 30 = 19.7 N

19.7 N down the ramp

Answer: Fg,parallel = 49.7 N downhill ∑F = 19.7 N down the ramp

RECALL

SECTION SUMMARY

Close your notes before you write.

Use the checklist to find what still needs practice.

Describe how to split the weight on a ramp, and how FN changes as the ramp gets steeper.

Fg,parallel = mg sin θ, Fg,perp = mg cos θ, FN = mg cos θ. Steeper ramp: smaller FN.

- ☐ I can write Fg,parallel = mg sin θ and Fg,perp = mg cos θ.
- ☐ I can find FN and the net force along a ramp.

Net Force and Acceleration

LEARNING TARGET I can use ∑F = ma to find acceleration, including on a ramp and in an elevator.

CALCULATING ACCELERATION

Slide 25

Given: m = 4 kg; ∑F = 12 N to the right. **Find:** a, and 1 N in base units.

WORK / show your reasoning and units

∑F = ma, so a = ∑F / m

a = 12 N / 4 kg

a = 3.0 m/s² right

1 N = 1 kg·m/s². a points in the direction of ∑F.

Answer: a = 3.0 m/s² right; 1 N = 1 kg·m/s²

A CAR ACCELERATES

Slide 26

Slide 27: worked on the board.

Given: m = 1500 kg; 4500 N forward; 1200 N drag (backward). Positive: forward. **Find:** ∑F and a.

WORK / show your reasoning and units

∑F = 4500 – 1200 = 3300 N forward

a = ∑F / m = 3300 / 1500

a = 2.2 m/s² forward

Answer: ∑F = 3300 N forward a = 2.2 m/s² forward

A SLED PULLED AT AN ANGLE

Slide 28

FN here is less than mg

Given: m = 20 kg; 60 N pull at 35° above horizontal; ignore friction. Positive: +x (right) and up.

Find: the acceleration and the normal force.

WORK / show your reasoning and units

∑Fx = 60 cos 35° = 49.1 N

a = ∑Fx / m = 49.1 / 20 = 2.46 m/s² right

Vertical: FN + 60 sin 35° – mg = 0

FN = 196 – 34.4 ≈ 162 N up

Answer: a = 2.46 m/s² right FN ≈ 162 N up

ACCELERATION DOWN A FRICTIONLESS RAMP

Slide 29

Given: m = 8 kg; frictionless 20° ramp. Positive: downhill. **Find:** a, showing why mass cancels.

WORK / show your reasoning and units

$\sum F = mg \sin \theta$

$ma = mg \sin \theta$

Mass cancels: $a = g \sin \theta = 9.8 \sin 20°$

$a = 3.35\ m/s^2$ downhill

Without friction, a does not depend on mass.

Answer: $a = 3.35\ m/s^2$ downhill

ELEVATORS AND APPARENT WEIGHT

Slide 30

Common mix-up: assuming the scale always reads mg.

Apparent weight: what a scale reads

The upward normal force, FN.

Elevator equation (up positive) and the a = 0 case

FN – mg = ma. At rest or constant velocity, a = 0, so FN = mg.

Scale reading: what changes it

Acceleration, not simply moving up or down.

ELEVATOR ACCELERATING UP

Slide 31

Slide 32: worked on the board.

Given: m = 70 kg; $a = 2.0\ m/s^2$ upward. Positive: up. **Find:** the scale reading.

WORK / show your reasoning and units

Weight: mg = 70(9.8) = 686 N

FN – mg = ma

FN = 686 + 70(2.0)

FN = 826 N up

Answer: FN = 826 N up

RECALL

SECTION SUMMARY

Close your notes before you write.

Use the checklist to find what still needs practice.

List the steps from a free-body diagram to an acceleration. Add how a scale reading changes in an accelerating elevator.

Choose a positive direction, add the forces to get ∑F, then a = ∑F / m in the direction of ∑F. The scale reads FN; FN – mg = ma, so FN changes when a ≠ 0.

- [ ] I can find a from ∑F = ma, with units and direction.
- [ ] I can find a scale reading with FN – mg = ma.

Friction

LEARNING TARGET I can tell static from kinetic friction and find friction, net force and acceleration on level ground and on a ramp.

STATIC AND KINETIC FRICTION · COMPARING COEFFICIENTS

Slides 34–35

Table values are approximate (OpenStax)

Friction: definition

Opposes sliding, or the tendency to slide, between surfaces.

Static vs. kinetic friction: equations

Static (not sliding): fs matches the needed force, up to fs,max = μsFN; at rest with a 20 N push, fs = 20 N opposite the push. Kinetic (sliding): fk ≈ μkFN, usually smaller.

μ: units, and what a higher value means

No units. At the same FN, higher μk means more sliding friction (shoes on ice 0.05; knee joint 0.015).

Extra notes:

FRICTION AT CONSTANT VELOCITY

Slide 36

Given: m = 40 kg; level floor; constant velocity; 120 N horizontal push. **Find:** FN, fk, and μk.

WORK / show your reasoning and units

Vertical balance: FN = mg = 40(9.8) = 392 N up

Constant velocity: a = 0, so fk = 120 N, opposite the push

μk = fk / FN = 120 / 392 ≈ 0.31 (no units)

Answer: FN = 392 N up fk = 120 N opposite the push μk ≈ 0.31

STATIC OR SLIDING: THE 25 KG BOX

Slide 37

Given: m = 25 kg, starts at rest; μs = 0.40; μk = 0.30; pushes of 90 N and 110 N. Positive: the push direction.

Find: whether the box moves for each push, and the acceleration for 110 N.

WORK / show your reasoning and units

FN = 25(9.8) = 245 N; fs,max = 0.40(245) = 98 N

90 N < 98 N: static friction is 90 N. The box stays at rest.

110 N > 98 N: the box slides. fk = 0.30(245) = 73.5 N

$\sum F = 110 - 73.5 = 36.5$ N; $a = 36.5 / 25 = 1.46$ m/s²

Answer: 90 N: stays at rest 110 N: slides, a = 1.46 m/s² in the push direction

ANGLED PULL WITH FRICTION

Slide 38

Given: m = 30 kg, sliding; 150 N rope at 30° above horizontal; μk = 0.25. Positive: +x (horizontal, toward the pull) and up.

Find: FN, fk, and a.

WORK / show your reasoning and units

FN = mg – F sin 30° = 294 – 75 = 219 N up

fk = 0.25(219) = 54.75 N, opposite the motion

∑Fx = 150 cos 30° – 54.75 = 75.15 N

a = 75.15 / 30 = 2.51 m/s² in +x

Answer: FN = 219 N up fk = 54.75 N opposite the motion a = 2.51 m/s² in +x

FRICTION ON AN INCLINE

Slide 39

Friction on a ramp: direction and equations

Up the ramp for a block sliding down. FN = mg cos θ; fs,max = μsFN; fk ≈ μkFN. Reverses if sliding reverses.

Condition to start sliding down

mg sin θ > μsFN

SLIDING DOWN THE RAMP: THE 10 KG BLOCK

Slide 40

Given: m = 10 kg, at rest; 35° ramp; μs = 0.50; μk = 0.40. Positive: downhill. **Find:** if it slides, and a.

WORK / show your reasoning and units

Fg,parallel = 10(9.8) sin 35° = 56.2 N; FN = 10(9.8) cos 35° = 80.3 N

fs,max = 0.50(80.3) = 40.1 N. 56.2 > 40.1, so it slides.

fk = 0.40(80.3) = 32.1 N

a = (56.2 – 32.1)/10 = 2.41 m/s² downhill

Answer: Slides: yes a = 2.41 m/s² downhill

RECALL

SECTION SUMMARY

Close your notes before you write.

Use the checklist to find what still needs practice.

Explain how to decide whether an object at rest starts to slide, and how to find its acceleration once it does.

Compare the push (or mg sin θ on a ramp) with fs,max = μsFN. If it is larger, it slides: use fk ≈ μkFN, find ∑F, then a = ∑F / m.

- ☐ I can tell static from kinetic friction and use fs,max = μsFN.
- ☐ I can find friction and acceleration on level ground and on a ramp.

Connected Objects

LEARNING TARGET I can find tension and acceleration for objects joined by cables, ropes and pulleys.

STREETLIGHT CABLE TENSION

Slide 42

Given: 20 kg streetlight at rest; left cable 30° above horizontal; right cable 60°; ignore cable mass. Positive: right and up.

Find: the tension in each support cable, and which carries more.

WORK / show your reasoning and units

At rest: $\sum F_x = 0$, $\sum F_y = 0$; $F_{down} = mg = 20(9.8) = 196$ N

Horizontal: $F_{TR} \cos 60° = F_{TL} \cos 30°$, so $F_{TL} = 0.577 F_{TR}$

Vertical: $F_{TL} \sin 30° + F_{TR} \sin 60° = 196$

$(0.577 F_{TR})(0.5) + 0.866 F_{TR} = 196$

$F_{TR} \approx 170$ N $F_{TL} = 98.0$ N, each along its cable, toward its support

The steeper cable carries more tension.

Answer: $F_{TL} = 98.0$ N $F_{TR} \approx 170$ N each along its cable, toward its support; the steeper cable carries more

IDEAL STRINGS AND PULLEYS

Slide 43

Ideal string and pulley

Massless, taut rope that does not stretch; massless, frictionless pulley.

Fixed-pulley rules and the method

Same tension T and same size of a for both objects. Find a from the whole system, then T from one object. Separate FBD for each object.

Extra notes:

ATWOOD MACHINE

Slide 44

Given: $m_1 = 5$ kg, $m_2 = 3$ kg, both hanging. Positive: the motion (5 kg down, 3 kg up). **Find:** a and T.

WORK / show your reasoning and units

$\sum F = (5 - 3)(9.8) = 19.6$ N

Whole system: $a = 19.6 / (5 + 3) = 2.45$ m/s² (5 kg down, 3 kg up)

Use the 3 kg mass: $T - 3g = 3a$

$T = 3(9.8 + 2.45) = 36.8$ N

Answer: $a = 2.45$ m/s² (5 kg down, 3 kg up) $T = 36.8$ N up on each mass

A BLOCK ON A FRICTIONLESS TABLE

Slide 45

Hanging mass accelerates down; table block, toward the pulley

Given: m_1 = 4 kg on the table; m_2 = 2 kg hanging. Positive: the motion. **Find:** a and T.

WORK / show your reasoning and units

Driving force: $\sum F = m_2 g = 2(9.8) = 19.6$ N

Whole system: $a = 19.6 / (4 + 2) = 3.27$ m/s² (2 kg down, 4 kg toward the pulley)

Table block: $T = m_1 a = 4(3.27) \approx 13.1$ N

Answer: a = 3.27 m/s² (2 kg down, 4 kg toward the pulley) T ≈ 13.1 N, toward the pulley on the table block, up on the hanging mass

THE SAME SYSTEM WITH FRICTION

Slide 46

Given: m_1 = 4 kg on the table, μk = 0.20; m_2 = 2 kg hanging. Positive: the motion. **Find:** fk, a, and T.

WORK / show your reasoning and units

$f_k = \mu k m_1 g = 0.20(4)(9.8) = 7.84$ N, away from the pulley

Whole system: $a = (19.6 - 7.84) / 6 = 1.96$ m/s² (2 kg down, 4 kg toward the pulley)

Table block: $T - f_k = m_1 a$

$T = 4(1.96) + 7.84 = 15.7$ N

Answer: fk = 7.84 N away from the pulley a = 1.96 m/s² (2 kg down, 4 kg toward the pulley) T = 15.7 N, toward the pulley on the table block, up on the hanging mass

CHALLENGE: INCLINE AND PULLEY

Slide 47

Given: m_1 = 6 kg on a frictionless 30° ramp; m_2 = 5 kg hanging. Positive: the motion (5 kg down, 6 kg up the ramp).

Find: a and T.

WORK / show your reasoning and units

Hanging: $5(9.8) = 49.0$ N; along ramp: $6(9.8) \sin 30° = 29.4$ N

Whole system: $a = (49.0 - 29.4) / (5 + 6) = 1.78$ m/s² (5 kg down, 6 kg up the ramp)

Hanging mass: $5g - T = 5a$

$T = 5(9.8 - 1.78) \approx 40.1$ N

Answer: a = 1.78 m/s² (5 kg down, 6 kg up the ramp) T ≈ 40.1 N, up the ramp on the 6 kg block, up on the 5 kg mass

RECALL

SECTION SUMMARY

Close your notes before you write.

Use the checklist to find what still needs practice.

Write the two-step method for objects joined by a rope over a pulley.

Whole system: a = (driving force – opposing forces) / total mass. Then ∑F = ma on one object gives T.

- [ ] I can solve a cable-tension problem with $\sum F_x = 0$ and $\sum F_y = 0$.
- [ ] I can find a and T for an Atwood machine and a table-and-pulley system.

Connect the ideas, check common mix-ups, and explain the physics in your own words.

S03.1 Newton's laws

Mass (kg) stays the same from place to place; weight, Fg = mg (N), depends on g. ∑F = 0: constant velocity.

Watch out: A moving object does not need a net force to keep moving.

S03.2 Free-body diagrams

One object, one labeled arrow per force. Add with signs; split an angled force into Fx = F cos θ and Fy = F sin θ.

Watch out: Third-law pairs act on two objects, so they never cancel on one FBD.

S03.3 Forces on an incline

Tilt the axes: Fg,parallel = mg sin θ, Fg,perp = mg cos θ. FN = mg cos θ (ramp, no other perpendicular force).

Watch out: Normal force does not always equal weight. A steeper ramp gives a smaller FN.

S03.4 Net force and acceleration

a = ∑F / m, in the direction of ∑F. Frictionless ramp: a = g sin θ. Elevator, up positive: FN – mg = ma.

Watch out: The scale reads FN, which changes with acceleration, not with moving up or down.

S03.5 Friction

Static friction matches the need, up to fs,max = μsFN. Sliding friction, fk ≈ μkFN, is usually smaller.

Watch out: Find FN first. An angled pull or a ramp changes it.

S03.6 Connected objects

Ideal rope: one tension T, one size of a. Find a from the whole system, then T from one object's ∑F = ma.

Watch out: Draw a separate FBD for each object.